DO NOW — RECALL

Modularity, functions & procedures

Name: _____ Class: _____ Date: _____

1. Which loop is most suitable when the number of repetitions is known in advance?

2. In C#, what does && mean, and when is the whole expression true?

3. What is the largest unsigned 8-bit denary value?

4. A program reads a mark, repeats through five scores and uses if to count passes. Name the three programming constructs present.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A for loop . It is commonly used for count-controlled iteration.

2. Model answer

&& means AND . The whole expression is true only when both Boolean expressions are true.

3. Model answer

255 Eight bits give 256 patterns, representing values 0–255.

4. Model answer

Sequence · iteration · branching/selection Instructions happen in order, a loop repeats code, and if chooses whether the pass counter changes.

By the end, you can…

• explain why programs are split into smaller modules;

• write and call a C# procedure using void ;

• write and call a C# function that returns a value.

Vocabulary — know these terms

Recall the term — Modularity

Recall the term — Subroutine

Recall the term — Method

Recall the term — Function

Recall the term — Procedure

Recall the term — Return value

Recall the term — Parameter

Recall the term — Argument

NEW LEARNING

Read, annotate and explain

Methods sit outside Main

```
public class Program
{
public static void Main()
{
Greeting("Aisha");
int answer = Triple(7);
}

public static void Greeting(string name)
{
Console.WriteLine("Hello " + name);
}

public static int Triple(int number)
{
return number * 3;
}
}
```

OCR language: a function returns a value. A procedure does not. In C#, both are methods.

Modularity breaks a problem into parts

A large program is easier to understand when separate jobs are placed in separate subroutines.

A procedure performs a job

A C# procedure-style method uses void because it does not return a value to the caller.

A function returns a value

The return type replaces void . A return statement sends the result back to the calling code.

Parameters receive data

A parameter is the named input in the method definition. An argument is the actual value supplied when the method is called.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Build an IsEven function

Choose the return type

```
public static bool IsEven(int number)
{
}
```

The result is true/false, so the return type is bool .

Use the remainder test

```
public static bool IsEven(int number)
{
    return number % 2 == 0;
}
```

The Boolean expression itself already produces true or false .

Use it from Main

```
public static void Main()
{
    bool result = IsEven(12);
    Console.WriteLine(result);
}
```

The argument 12 is received by the parameter number .

Use contrasting values

```
IsEven(12) → True
IsEven(7)  → False
IsEven(0)  → True
```

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY: Greeting procedure

Create a method called Greeting that receives a name and displays Hello name .

Method / working:

Guided 2 — TRY: Square function

Create a method that receives an integer and returns its square.

```
public static ? Square(int number)
{
    ?
}
```

Method / working:

Guided 3 — TRY: Refactor repeated code

A program prints the same separator in several places: Move that repeated job into one method.

Method / working:

Guided 4 — TRY: Function or procedure?

OCR asks students to distinguish the two. State the difference precisely.

Method / working:

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

DisplayTitle procedure
Create a procedure-style method called DisplayTitle() .
It should output OCR H446 Computer Science .
Call it from Main() twice.

Question 2

DoubleIt function
Create DoubleIt(int number) that returns twice the value supplied.
Call the function with 12 , store the returned value, then output it.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

IsPass function
Create IsPass(int mark) .
Return true when the mark is at least 50; otherwise return false .
Call it from Main() with at least two different marks.

Question 4

Refactor a program into modules
Write a program that asks for two integers and then:
1. calls Add(int first, int second) and outputs the returned total;
2. calls ShowLine() before and after the result.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

A function returns a value to its caller.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

Function or procedure?

Guided 2 — CHECK

Complete the method
CODE:
public static ? Square(int number)
{
?
}
CODE:
public static int Square(int number)
{
return number * number;
}

Guided 3 — CHECK

Design the module

Guided 4 — CHECK

Complete both statements